

SingleRAN

Ethernet OAM Feature Parameter Description

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1 Change History

This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes
Changes in functions and their corresponding parameters
- Editorial changes
Improvements or revisions to the documentation

1.1 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 01 (2025-03-10).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added incompatibility of BTS5929R with Ethernet OAM. For details, see 4.3.3 Hardware and 5.4.3 Hardware .	None	LTE FDD	3900 and 5900 series base stations (macro base stations) excluding the BTS5929R

- Editorial changes
Revised descriptions in this document.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
GSM	GBFD-118630	Ethernet OAM	• 4 Ethernet OAM (IEEE 802.3ah) • 5 Ethernet OAM (IEEE 802.1ag/ITU-T Y.1731)
UMTS	WRFD-050425	Ethernet OAM	
LTE FDD	LOFD-003004	Ethernet OAM	
LTE TDD	TDLOFD-003004	Ethernet OAM	
NB-IoT	MLOFD-003004	Ethernet OAM	
NR	FOFD-010060	Transmission Network Detection and Reliability Improvement	

3 Overview

3.1 Introduction

With the advent of Internet Protocol (IP) transformation in telecom operator networks, Ethernet bearer networks are widely used. To guarantee the reliability and availability of Ethernet services, the Ethernet OAM feature is introduced.

This feature provides detection, notification, verification, and locating functions for detectable hard faults (such as link disconnection) or undetectable soft faults (such as damages in memory bridges).

From this version onwards, Ethernet OAM supports the following three protocols:

- Point-to-point IEEE 802.3ah
- End-to-end IEEE 802.1ag
- End-to-end ITU-T Y.1731

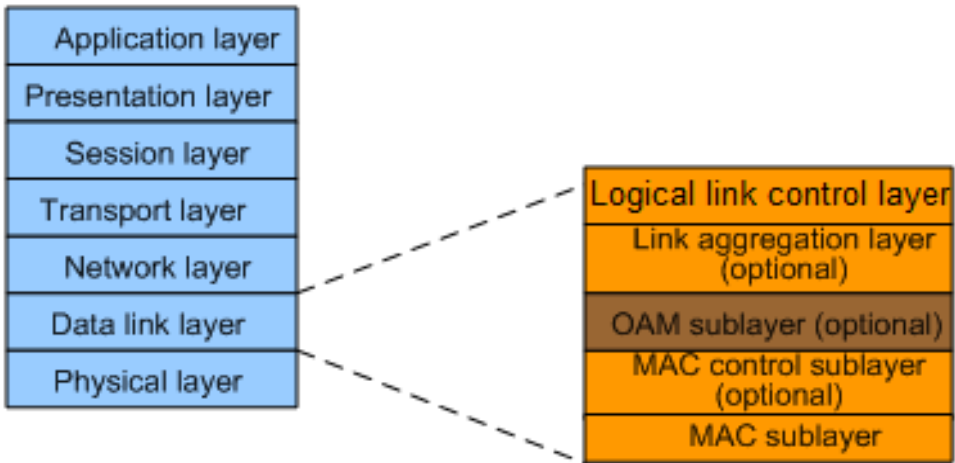
NOTE

- Point-to-point Ethernet OAM IEEE 802.3ah: Two pieces of equipment are directly connected and there cannot be any Ethernet switch equipment between them.
- End-to-end Ethernet OAM IEEE 802.1ag or Ethernet OAM ITU-T Y.1731: Two pieces of equipment are not directly connected and have layer 2 switches between them.

3.2 Architecture

The Ethernet OAM feature detects and reports network status at the data link layer. [Figure 3-1](#) shows the position of the Ethernet OAM protocols in the OSI model.

Figure 3-1 Position of Ethernet OAM protocols in the OSI model



Ethernet OAM protocols are functional between the Media Access Control (MAC) sublayer and the logical link control (LLC) layer. Therefore, other protocol sublayers at the Ethernet data link layer and other OSI layers are not affected.

4 Ethernet OAM (IEEE 802.3ah)

4.1 Principles

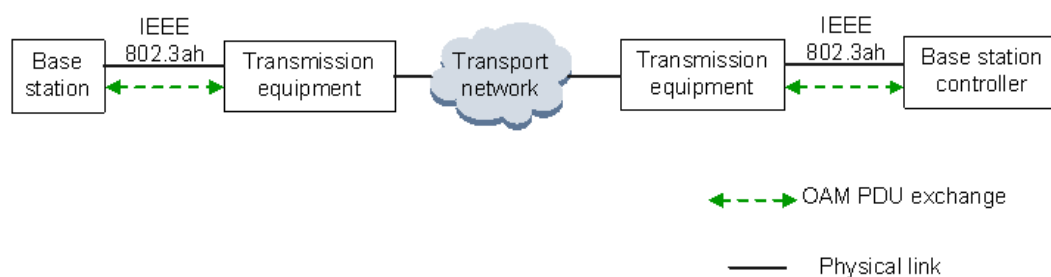
4.1.1 Introduction

Ethernet OAM (IEEE 802.3ah) is a point-to-point Ethernet OAM protocol used to implement the Ethernet in the First Mile (EFM) OAM function and effectively monitor the link status.

The EFM OAM function must be enabled for two pieces of equipment that are directly connected during point-point Ethernet OAM services. The function detects faults on Ethernet links by using the automatic discovery, fault detection, link monitoring, and remote loopback mechanisms.

Figure 4-1 shows an example of Ethernet OAM (IEEE 802.3ah) deployed between a base station (or base station controller) and transmission equipment that are directly connected through the Ethernet port. OAM protocol data units (PDUs) are exchanged between them to detect the link status. In this example, the remote end transmission equipment must support IEEE 802.3ah.

Figure 4-1 Schematic diagram of Ethernet OAM (IEEE 802.3ah)



4.1.2 Basic Concepts

Ethernet OAM (IEEE 802.3ah) supports active and passive modes.

- In active mode, the local end proactively sends negotiation packets to the remote end.

- In passive mode, the local end waits for negotiation packets from the remote end.

It is recommended that Ethernet OAM (IEEE 802.3ah) work in active mode. In applications, only when both ends at a point-to-point link work in passive mode, neither end initiates the automatic discovery procedure. The automatic discovery procedure is also known as OAM discovery. To implement OAM functions, automatic discovery must be enabled first. OAM clients in different modes have different OAM support capabilities. For details, see [Table 4-1](#).

Table 4-1 OAM client capabilities in different modes

OAM Capability	Active Mode	Passive Mode
Initiating automatic discovery	Yes	No
Responding to automatic discovery initiation	Yes	Yes
Sending Information OAM PDUs	Yes	Yes
Sending Event Notification OAM PDUs	Yes	Yes
Sending Variable Request OAM PDUs	Yes	No
Sending Variable Response OAM PDUs	Yes	Yes
Sending Loopback Control OAM PDUs	Yes	No
Responding to Loopback Control OAM PDUs	Yes	Yes
Sending Organization Specific OAM PDUs	Yes	Yes

4.1.3 Application Scenarios

In IEEE 802.3ah-based networking, the local end must be directly connected to the peer end. As shown in [Figure 4-2](#), this scenario is used to manage the Ethernet link between the base station or base station controller and the transmission device.

Figure 4-2 Network topology with Ethernet OAM (IEEE 802.3ah) enabled



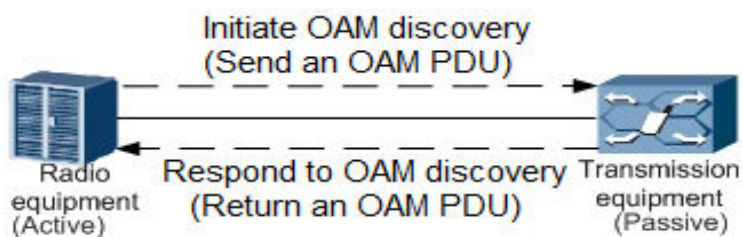
4.1.4 Technical Description

IEEE 802.3ah defines four basic functions: automatic discovery, fault detection, link monitoring, and remote loopback. Automatic discovery refers to the OAM session establishment and the exchange procedure. It is the prerequisite for the other functions. Link monitoring, fault detection, and remote loopback are used to monitor network health and quickly locate link failures.

Automatic Discovery

Automatic discovery is the first phase for the implementation of Ethernet OAM (IEEE 802.3ah) and is the prerequisite for other OAM functions. This function provides a process of establishing and interacting OAM sessions. For example, if the radio equipment works in active mode and the transmission equipment works in passive mode, the two ends exchange OAM PDUs to obtain the OAM support capabilities of their respective peer ends, as shown in [Figure 4-3](#).

Figure 4-3 Automatic discovery process



Fault Detection

- If the local end is faulty, the local OAM entity notifies the peer OAM entity of the fault by sending an OAM PDU that contains the fault information.
- If the peer end is faulty, the peer OAM entity notifies the local OAM entity of the fault by sending an OAM PDU that contains the fault information.

Link Monitoring

Link monitoring is used to monitor bit errors, such as errored frames and errored signals on a link. When the bit error rate (BER) exceeds a specified threshold, the local OAM entity notifies the peer OAM entity of the corresponding bit error event by sending an Event Notification OAM PDU. Then, the peer OAM entity reports the link fault.

To measure the performance from different perspectives, the link monitoring function is divided into three event monitoring periods:

- Errored frame event monitoring period: refers to the period in which errored frames are measured once. In this period, the number of detected errored frames is greater than or equal to the specified errored frame event threshold. The related parameters for GSM/UMTS/LTE/NR are described in the following table.

NE	Parameter Name	Parameter ID
BSC/RNC	Err Frame Event Prid[s]	EFMAH.ERFAMEventPrid
	Err Frame Event Para	EFMAH.ERFAMEventPara
GBTS	Err Frame Event Prid[s]	EFMAH.ERFAMEventPrid
	Err Frame Event Para	EFMAH.ERFAMEventPara
eGBTS/NodeB/ eNodeB/gNodeB	Error Frame Event Duration	ETHOAM3AH.ERRFRMEVTPRD (LTE eNodeB, 5G gNodeB)/ ETH3AH.ERRFRMEVTPRD (LTE eNodeB, 5G gNodeB)
	Error Frame Event Threshold	ETHOAM3AH.ERRFRMEVTT HD (LTE eNodeB, 5G gNodeB)/ ETH3AH.ERRFRMEVTT HD (LTE eNodeB, 5G gNodeB)

- Errored frame seconds event monitoring period: In this period, the number of detected errored frame seconds is greater than or equal to the specified errored frame seconds summary event threshold. An errored frame seconds event indicates that at least one errored frame is detected in a second. The related parameters for GSM/UMTS/LTE/NR are described in the following table.

NE	Parameter Name	Parameter ID
BSC/RNC	Err Frame Sec Event Prid[s]	EFMAH.ERFAMSCDEventPrid

NE	Parameter Name	Parameter ID
	Err Frame Sec Event Para[s]	EFMAH.ERFAMSCDEventPara
GBTS	Err Frame Sec Event Prid[s]	EFMAH.ERFAMSCDEventPrid
	Err Frame Sec Event Para[s]	EFMAH.ERFAMSCDEventPara
eGBTS/NodeB/ eNodeB/gNodeB	Error Frame Seconds Summary Event Duration	ETH3AH.ERRFRMSCDEVTPRD (LTE eNodeB, 5G gNodeB)/ ETHOAM3AH.ERRFRMSCDEVTPRD (LTE eNodeB, 5G gNodeB)
	Error Frame Seconds Summary Event Threshold	ETH3AH.ERRFRMSCDEVTT HD (LTE eNodeB, 5G gNodeB)/ ETHOAM3AH.ERRFRMSCDEVTT HD (LTE eNodeB, 5G gNodeB)

- Errored frame period event monitoring period In this period, the number of detected received errored frames is greater than or equal to the specified threshold for the received errored frame cycle events. The related parameters for GSM/UMTS/LTE/NR are described in the following table.

 NOTE

The BSC/RNC and GBTS support this function. The NodeB/eNodeB/gNodeB/eGBTS does not support this function.

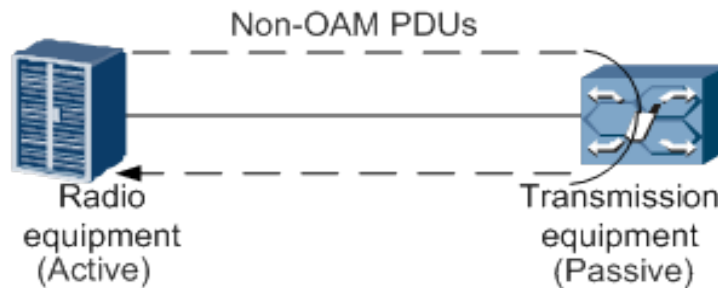
NE	Parameter Name	Parameter ID
BSC/RNC	Err Frame Cyc Event Prid	EFMAH.ERFAMPRDEventPrid
	Err Frame Cyc Event Para	EFMAH.ERFAMPRDEventPara
GBTS	Err Frame Cyc Event Prid	EFMAH.ERFAMPRDEventPrid
	Err Frame Cyc Event Para	EFMAH.ERFAMPRDEventPara

Remote Loopback

When the local OAM entity (wireless equipment) sends packets to the peer OAM entity (transmission equipment), the peer OAM entity loops back packets other than OAM PDUs. Remote loopback can be used to locate link faults and test link quality.

The local OAM entity sends a Loopback Control OAM PDU to the peer OAM entity, requesting the peer OAM entity to enter the loopback state. As shown in [Figure 4-4](#), after the peer OAM entity enters the loopback state, the local end can send test packets to the peer end. By comparing the number of sent packets with the number of received packets, you can calculate the communication quality parameters, such as the packet loss rate on the current link.

Figure 4-4 Remote loopback



Only the port in active mode can initiate a remote loopback and both ports must be detectable. The remote loopback function is implemented as follows:

1. The local end sends a loopback request message to the peer end and waits for a response.
2. After receiving the loopback request message, the peer port responds to the local end with a loopback response message and enters the loopback state.
3. If the local end receives the loopback response message within a certain test period, the local end enters the loopback state. If the local end does not receive the loopback response message within the test period, the local end sends a loopback request message to the peer end again. The radio equipment can send the command three consecutive times at most.
4. When the local end needs to stop the remote loopback, the local end sends a loopback cancellation message to the peer end. After receiving the message, the peer end exits the loopback state.

NOTE

- When the working mode of a base station is set to **PASSIVE**, only the packets of which the destination MAC address is the MAC address of the local port of the base station can be looped back.
- Remote loopback is not supported by the BSC6960.

4.2 Network Analysis

4.2.1 Benefits

Ethernet OAM helps operators improve management and maintenance of the Ethernet, guarantee network reliability, and reduce operating expense (OPEX).

4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

Ethernet OAM protocols are slow-rate protocols and therefore can provide low data rate functions, such as the ping function. In addition, Ethernet OAM protocols occupy low bandwidth and therefore do not affect the data flow of other services. If the CCM function is enabled, packets must be sent at an interval of at least one second. This is because sending packets at a very short interval affects the performance of transport networks. When the remote loopback function provided by Ethernet OAM (IEEE 802.3ah) is enabled, services carried on a base station are interrupted.

4.3 Requirements

4.3.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
UMTS	WRFD-050425	Ethernet OAM	LQW9ETOAM01	per NodeB
LTE FDD	LOFD-003004	Ethernet OAM	LT1S00EOAM00	per eNodeB
LTE TDD	TDLOFD-003004	Ethernet OAM	LT1ST0EOAM00	per eNodeB
NB-IoT	MLOFD-003004	Ethernet OAM	ML1S00EOAM00	per eNodeB
NR	FOFD-010060	Transmission Network Detection and Reliability Improvement	NR0S0TNDER00	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
GSM	Abis over IP	None	<i>IPv4 Transmission</i>
GSM	A over IP	None	
GSM	Gb over IP	None	
UMTS	IP Transmission Introduction on Iub Interface	None	

4.3.2.2 Mutually Exclusive Functions

None

4.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
GSM UMTS	Link Aggregation	None	<i>IPv4 Transmission</i>	Ethernet trunk member ports support Ethernet OAM (IEEE 802.3ah), but Ethernet trunk ports do not.
LTE FDD	Ethernet Link Aggregation	None		
LTE TDD	Ethernet Link Aggregation (IEEE 802.3ad)	None		
CloT	Ethernet Link Aggregation (IEEE 802.3ad)	None		
NR FDD Low-frequency NR TDD	Transmission Network Detection and Reliability Improvement - Ethernet Link Aggregation	None		
LTE FDD LTE TDD	S1 and X2 over IPv6	None	<i>IPv6 Transmission</i>	Ethernet OAM (IEEE 802.3ah) does not support IPv6 transmission.

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	S1 over IPv6	None	<i>IPv6 Transmission</i>	Ethernet OAM (IEEE 802.3ah) does not support IPv6 transmission.
NR FDD Low-frequency NR TDD	IPv6	None	<i>IPv6 Transmission</i>	Ethernet OAM (IEEE 802.3ah) does not support IPv6 transmission.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

The following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
- DBS3900 LampSite and DBS5900 LampSite

Boards

RAT	NE	Board	Supported
GSM	BSC	FG2a/FG2c/FG2d/FG2e/GOUa/GOUc/GOUd/EXOUa/EXOUb/CMPU	Yes
GSM	GBTS	GTMUc	Yes
GSM	eGBTS	UMPT	Yes
UMTS	RNC	FG2a/FG2c/FG2d/FG2e/GOUa/GOUc/GOUd/EXOUa/EXOUb/CMPU	Yes
	NodeB	UMPT/UTRPc	Yes
LTE	eNodeB	UMPT/UMDU/BookBBU5901/BookBBU5901a/BBU5900C/UTRPc	Yes
NR	gNodeB	UMPTe/UMPTg/UMPTga/UMPTga/UMDU/BookBBU5901/BookBBU5901a/BBU5900C	Yes

RF Modules

N/A

Transmission Equipment

The transmission equipment directly connected to the peer end must support the IEEE 802.3ah function.

4.3.4 Others

None

4.4 Operation and Maintenance

4.4.1 When to Use

Before Ethernet OAM is enabled, the following conditions must be met:

- Both the base station or base station controller and the transport network are carried on a Layer 2 Ethernet.
- The transmission equipment supports Ethernet OAM.

Ethernet OAM (IEEE 802.3ah) is an effective link monitoring and fault detection mechanism during point-to-point Ethernet OAM (only for the base station side). It is recommended that Ethernet OAM (IEEE 802.3ah) be deployed for a base station or base station controller and adjacent transmission equipment that supports IEEE 802.3ah.

In base station co-transmission scenarios, Ethernet OAM is deployed only for a base station that is directly connected to the transport network. Ethernet OAM is deployed only for one RAT of a base station if multiple RATs of the base station are directly connected to the transport network.

4.4.2 Data Configuration

4.4.2.1 Data Preparation

To deploy Ethernet OAM, collect the following information:

- Whether the transmission equipment supports Ethernet OAM
- IEEE 802.1ag version (Draft 7 or standard version) supported by the transmission equipment during Ethernet OAM (IEEE 802.1ag) deployment
- Whether the transmission equipment supports ITU-T Y.1731 (2008) during Ethernet OAM (ITU-T Y.1731) deployment

The protocol version supported by the transmission equipment must be the same as that supported by the radio equipment.

The following table describes the versions for IEEE 802.1ag and ITU-T Y.1731 protocols supported by the radio equipment.

Protocol	Base Station	Base Station Controller
IEEE 802.1ag NOTE IEEE 802.1ag of the Draft version is incompatible with IEEE 802.1ag of the 2007 standard version. The two ends must use the same protocol version.	In versions from GBSS9.0/RAN9.0 to GBSS13.0/RAN13.0, the base station supports only IEEE 802.1ag (Draft 7). In GBSS14.0/RAN14.0/eRAN3.0/SRAN7.0 and later versions, the base station supports only the standard version.	The base station controller supports both the Draft 7 and standard version.
ITU-T Y.1731	The base station supports ITU-T Y.1731 (2008).	The base station controller does not support ITU-T Y.1731.

When deploying Ethernet OAM (IEEE 802.1ag) or Ethernet OAM (ITU-T Y.1731), plan the following data for the base station (or base station controller) and transmission equipment:

- Maintenance domain (MD): The MD is invalid during Ethernet OAM (ITU-T Y.1731) deployment.
- Maintenance association (MA): The MA is replaced with the MEG ID during Ethernet OAM (ITU-T Y.1731) deployment.
- Maintenance association end point (MEP)/Remote maintenance association end point (RMEP)
- VLAN

GSM

[Table 4-2](#) and [Table 4-3](#) list the data to prepare for activating Ethernet OAM (IEEE 802.3ah) on the BSC and GBTS, respectively. The data to prepare for activating Ethernet OAM (IEEE 802.3ah) on the eGBTS side is the same as that on the NodeB side. For details, see [Table 4-4](#).

Table 4-2 Parameters used for activation on the BSC side

Parameter Name	Parameter ID	Setting Notes
PDU Send Prid[ms]	EFMAH.PDUPrid	N/A
PDU Pkt Len[Byte]	EFMAH.PDUSize	N/A
Err Frame Event Prid[s]	EFMAH.ERFAMEventPrid	N/A
Err Frame Event Para	EFMAH.ERFAMEventPara	N/A

Parameter Name	Parameter ID	Setting Notes
Err Frame Cyc Event Prid	EFMAH.ERFAMPRDEventPrid	N/A
Err Frame Cyc Event Para	EFMAH.ERFAMPRDEventPara	N/A
Err Frame Sec Event Prid[s]	EFMAH.ERFAMSCDEventPrid	N/A
Err Frame Sec Event Para[s]	EFMAH.ERFAMSCDEventPara	N/A

Table 4-3 Parameters used for activation on the GBTS side

Parameter Name	Parameter ID	Setting Notes
PDU Send Prid	BTSETHOAMAH.PDUPrid	N/A
PDU Packet Len	BTSETHOAMAH.PDUSize	N/A
Err Frame Event Prid	BTSETHOAMAH.ERFAMPRID	N/A
Err Frame Event Para	BTSETHOAMAH.ERFAMPARA	N/A
Err Frame Cyc Event Prid	BTSETHOAMAH.ERFAMPRDPRID	N/A
Err Frame Cyc Event Para	BTSETHOAMAH.ERFAMPRDPARA	N/A
Err Frame Sec Event Prid	BTSETHOAMAH.ERFAMSCDPRID	N/A
Err Frame Sec Event Para	BTSETHOAMAH.ERFAMSCDPARA	N/A

UMTS

The data to prepare for activating this feature on the RNC side is the same as that on the BSC side. For details, see [Table 4-2](#). [Table 4-4](#) lists the data to prepare for activating this feature on the NodeB side.

Table 4-4 Parameters used for activation on the NodeB side

Parameter Name	Parameter ID	Setting Notes
Work Mode	ETHOAM3AH.WORKMODE (LTE eNodeB, 5G gNodeB)/ ETH3AH.WORKMODE (LTE eNodeB, 5G gNodeB)	N/A
Detection Period	ETHOAM3AH.DTPRD (LTE eNodeB, 5G gNodeB)/ ETH3AH.DTPRD (LTE eNodeB, 5G gNodeB)	N/A
PDU Size	ETHOAM3AH.PDUSIZE (LTE eNodeB, 5G gNodeB)/ ETH3AH.PDUSIZE (LTE eNodeB, 5G gNodeB)	N/A
Error Frame Event Duration	ETHOAM3AH.ERRFRMEVTPRD (LTE eNodeB, 5G gNodeB)/ ETH3AH.ERRFRMEVTPRD (LTE eNodeB, 5G gNodeB)	N/A
Error Frame Event Threshold	ETHOAM3AH.ERRFRMEVTTHD (LTE eNodeB, 5G gNodeB)/ ETH3AH.ERRFRMEVTTHD (LTE eNodeB, 5G gNodeB)	N/A
Error Frame Seconds Summary Event Duration	ETHOAM3AH.ERRFRMSCDEVTPRD (LTE eNodeB, 5G gNodeB)/ ETH3AH.ERRFRMSCDEVTPRD (LTE eNodeB, 5G gNodeB)	N/A
Error Frame Seconds Summary Event Threshold	ETHOAM3AH.ERRFRMSCDEVTTHD (LTE eNodeB, 5G gNodeB)/ ETH3AH.ERRFRMSCDEVTTHD (LTE eNodeB, 5G gNodeB)	N/A
NOTE • ETHOAM3AH is used in the old transmission configuration model. • ETH3AH is used in the new transmission configuration model.		

LTE&NR

The data to prepare on the eNodeB/gNodeB side is the same as that on the NodeB side. For details, see [Table 4-4](#).

4.4.2.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

GSM (on the GBSC Side)

When the BSC6900/BSC6910 is used:

```
ACT EFMAH: SRN=0, SN=18, PN=0;
```

When the BSC6960 is used:

```
ADD ETH3AH: ETH0AM3AHID=0, ETHPORTID=0, WORKMODE=ACTIVE, PDUSIZE=128, ERRFRMEVTPRD=1, ERRFRMEVTTHD=1, ERRFRMSCDEVTPRD=60, ERRFRMSCDEVTTHD=1;
```

On the GBTS Side

```
ACT BTSETHOAMAH: IDTYPE=BYID, BTSID=7, PN=0;
```

The activation command examples on the eGBTS side are the same as those on the NodeB side. For details, see "UMTS (on the NodeB Side)" in "Activation Command Examples."

UMTS (on the NodeB Side)

When the old transmission configuration model is used
(**GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**):

```
ACT ETHOAM3AH: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=ETH, PN=0, WORKMODE=ACTIVE, PDUSIZE=128, ERRFRMEVTPRD=1, ERRFRMEVTTHD=1, ERRFRMSCDEVTPRD=60, ERRFRMSCDEVTTHD=1;
```

When the new transmission configuration model is used
(**GTRANSPARA.TRANS CFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**):

```
ADD ETH3AH: ETH0AM3AHID=0, ETHPORTID=0, WORKMODE=ACTIVE, PDUSIZE=128, ERRFRMEVTPRD=1, ERRFRMEVTTHD=1, ERRFRMSCDEVTPRD=60, ERRFRMSCDEVTTHD=1;
```

The activation command examples on the RNC side are the same as those on the BSC side. For details, see "GSM (on the GBSC Side)" in "Activation Command Examples."

LTE/NR

The activation command examples on the eNodeB/gNodeB side are the same as those on the NodeB side. For details, see "UMTS (on the NodeB Side)" in "Activation Command Examples."

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

GSM (on the GBSC Side)

When the BSC6900/6910 is used:

DEA EFMAH: SRN=0,SN=18,PN=0;

When the BSC6960 is used:

RMV ETH3AH: ETH0AM3AHID=0;

GSM (on the GBTS Side)

DEA BTSETHOAMAH: IDTYPE=BYID, BTSID=7, PN=0;

The deactivation command examples on the eGBTS side are the same as those on the NodeB side. For details, see "UMTS (on the NodeB Side)" in "Deactivation Command Examples."

UMTS (on the NodeB Side)

When **GTRANSPARA.TRANSCFGMODE (5G gNodeB, LTE eNodeB)** is set to **OLD**:

DEA ETHOAM3AH: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PT=ETH, PN=0;

When **GTRANSPARA.TRANSCFGMODE (5G gNodeB, LTE eNodeB)** is set to **NEW**:

RMV ETH3AH: ETH0AM3AHID=0;

The deactivation command examples on the RNC side are the same as those on the BSC side. For details, see "GSM (on the GBSC Side)" in "Deactivation Command Examples."

LTE/NR

The deactivation command examples on the eNodeB/gNodeB side are the same as those on the NodeB side. For details, see "UMTS (on the NodeB Side)" in "Deactivation Command Examples."

4.4.3 Activation Verification

GSM

On the BSC side, run the **STR EFMLOOPTST** command and then run the **DSP EFMLOOPTST** command to query the value of **Information of ETHOAM remote loopback**. If the value is **EthOam 3ah Detect Status**, Ethernet OAM (IEEE 802.3ah) has been activated on the port.

On the GBTS side, run the **LST BTSETHOAMAH** command to check whether the Ethernet OAM (IEEE 802.3ah) has been configured.

The activation verification using MML commands on the eGBTS side is the same as that on the NodeB side. For details, see "On the NodeB Side" in [UMTS](#).

UMTS

On the RNC Side

The activation verification using MML commands on the RNC side is the same as that on the BSC side. For details, see "On the BSC Side" in [GSM](#).

On the NodeB Side

Run either of the following commands accordingly to query the status of Ethernet OAM (IEEE 802.3ah). If the value of **Current State** is **Discover Phase**, Ethernet OAM (IEEE 802.3ah) has been activated on the port.

- **DSP ETHOAM3AH** when the old transmission configuration model is used
- **DSP ETH3AH** when the new transmission configuration model is used

LTE/NR

The activation verification using MML commands on the eNodeB side is the same as that on the NodeB side. For details, see "On the NodeB Side" in [UMTS](#).

4.4.4 Network Monitoring

None

5 Ethernet OAM (IEEE 802.1ag/ITU-T Y.1731)

5.1 Principles (IEEE 802.1ag)

5.1.1 Introduction

Ethernet OAM (IEEE 802.1ag) performs the connectivity fault management (CFM) function specified in the IEEE 802.1ag protocol, which is an end-to-end Ethernet OAM protocol. The CFM function is a network-level Ethernet OAM technology and consists of continuity check, loopback, and linktrace functions. With the CFM function, the connectivity faults on the entire network can be detected and located.

If the base station or base station controller detects Ethernet faults or undesirable network performance using Ethernet OAM (IEEE 802.1ag), it initiates rerouting, port switchover, or board switchover accordingly. This ensures normal connection of Ethernet links.

5.1.2 Basic Concepts

Ethernet OAM (IEEE 802.1ag) reports the status of the data link layer and therefore effectively monitors and manages the network. To use this function, the local and peer devices must support IEEE802.1ag.

Ethernet OAM (IEEE 802.1ag) can be used to maintain end-to-end Ethernet links. To support fault detection and fault isolation, IEEE 802.1ag defines the maintenance domain (MD), maintenance association (MA), and maintenance point (MP). MPs include maintenance association end points (MEPs) and maintenance association intermediate points (MIPs).

Ethernet OAM (IEEE 802.1ag) is deployed based on services to implement end-to-end detection and perform segment-by-segment detection on each network segment that a service flow passes through on a network.

- Maintenance domain (MD)
The MD denotes a network or a part of a network in which the CFM is implemented. An MD is usually managed by an Internet Service Provider (ISP)

or an operator. The MD name is specified by the **MDNAME (LTE eNodeB, 5G gNodeB)** parameter.

MDs can be managed on a per level basis. The level of an MD is specified by the **Level (LTE eNodeB, 5G gNodeB)** parameter. A high-level MD can span low-level MDs, but a low-level MD cannot span high-level MDs. This level-based MD management allows the same service flow to be maintained segment by segment and also applies to different service flows.

Ethernet OAM (IEEE 802.1ag) supports eight MD levels, ranging from 0 to 7. Level 7 is the highest MD level, and level 0 is the lowest.

- Maintenance association (MA)

An MA is a part of an MD. An MD can be divided into one or more MAs. The continuity of each MA can be checked using the CFM function.

In networks of operators, one Virtual Local Area Network (VLAN) generally corresponds to one service instance and one MA. Therefore, the continuity check of service instances can be performed on a per MA basis.

- Maintenance association end point (MEP)

MEPs are end points of MAs. A MEP on a wireless device running IEEE 802.1ag is called a local MEP, and MEPs on other devices in the same MA are called remote maintenance association end points (RMEPs) for the local device.

- Maintenance association intermediate point (MIP)

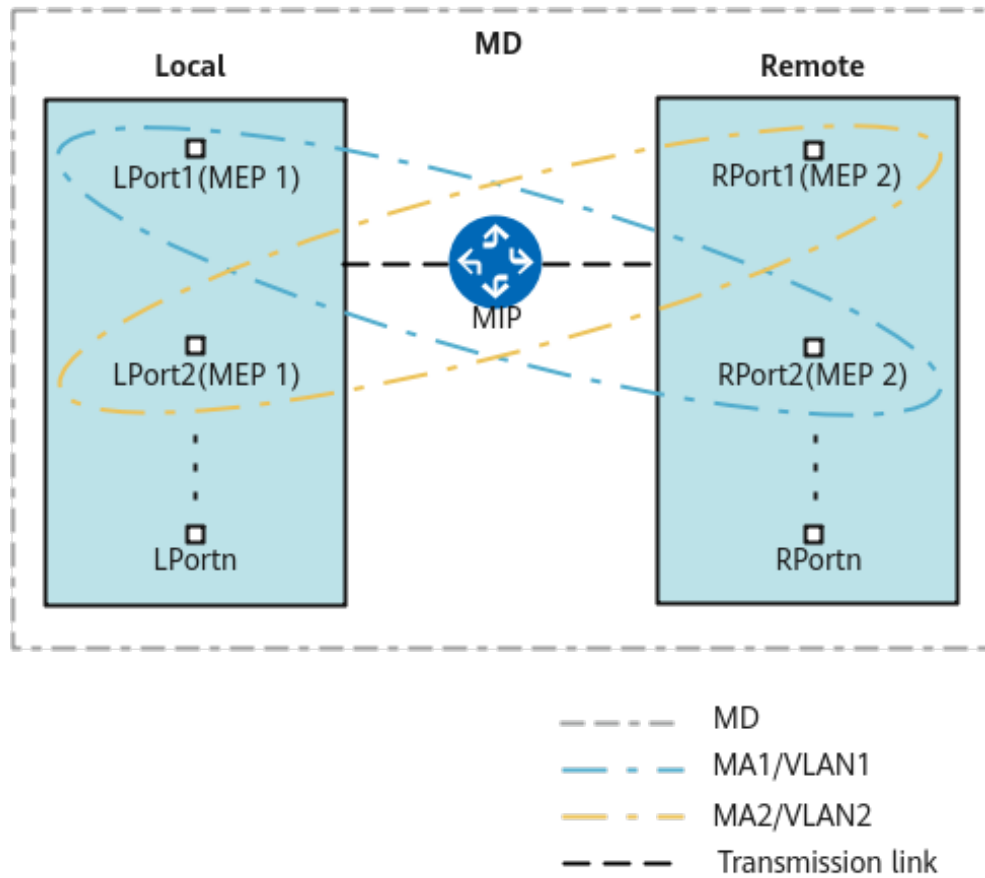
MIPs are intermediate points of MAs.

All OAM operations are initiated by MEPs. MIPs do not actively initiate any OAM operation or send any OAM packets.

Figure 5-1 shows the relationship between the MD, MA, MEP, MIP, and VLAN. The local end and remote end are configured in the same MD. In addition, two MAs, MA1 and MA2, are created in the MD. MEP 1 configured on LPort1 at the local end and MEP 2 configured on RPort2 at the remote end are in the same MA, MA1. MEP 1 configured on LPort2 at the local end and MEP 2 configured on RPort1 at the remote end are in the same MA, MA2. MEPIDs in different MAs can be the same.

In this example, LPort1 and LPort2 serve as local MEPs, and RPort1 and RPort2 serve as remote MEPs.

Figure 5-1 Relationship among MD, MA, MEP, MIP, and VLAN



NOTE

- Radio equipment can work as an MEP but cannot work as an MIP.
- If the radio equipment is an MEP of IEEE 802.1ag/ITU-T Y.1731 and **CFMMA.BNDVLAN (LTE eNodeB, 5G gNodeB)** is set to **Yes**, the intermediate switch cannot be set to the access mode. Otherwise, the ITU-T Y.1731 measurement is abnormal.

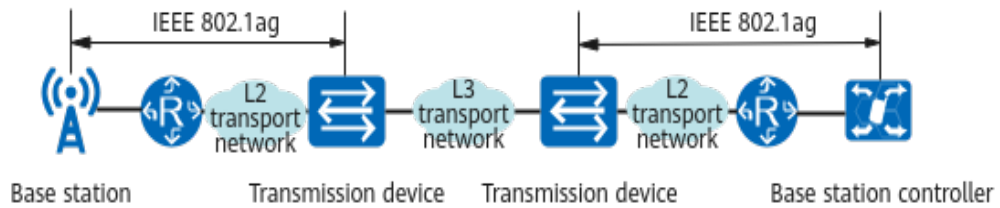
The MEPs at different MD levels process OAM packets as follows:

- MEPs transparently transmit the OAM packets from a higher MD level.
- MEPs discard the OAM packets from a lower MD level.
- MEPs respond to or terminate the OAM packets from the same MD level based on the OAM packet type.

5.1.3 Application Scenarios

Figure 5-2 shows the network topology when Ethernet OAM (IEEE 802.1ag) is used. Ethernet OAM (IEEE 802.1ag) is deployed between the GBTS/NodeB/eNodeB/gNodeB/eGBTS and transmission equipment, and between the BSC/RNC and transmission equipment.

Figure 5-2 Network topology with Ethernet OAM (IEEE 802.1ag) enabled



5.1.4 Technical Description

Ethernet OAM (IEEE 802.1ag) implements end-to-end Ethernet maintenance and detection using the following functions:

- Continuity check (CC)

CC is a basic function and is used to check the continuity of Ethernet links between MEPs on two pieces of Ethernet equipment.

When this function is enabled, each MEP in an MD periodically sends continuity check messages (CCMs) and all MEPs in the same MD can receive the CCM packets.

However, the MEPs do not need to respond to the received CCM packets. After receiving a CCM packet from the remote MEP, the local MEP waits for a second CCM packet from the remote MEP within a specific period. When the period elapses but the local MEP does not receive any CCM packet from the remote MEP, the local MEP determines that a continuity fault occurs between two ends. The specific period equals three times the interval for sending CCM packets from the remote MEP. In most cases, the local MEP uses the same interval as the remote MEP.

- Loopback (LB)

LB is used to detect the bidirectional continuity between the local end and remote end.

The local MEP sends a Loopback Message which contains the MAC address or MEP ID of the destination MP (MEP or MIP) and starts a timer.

- If the destination MP responds with a Loopback Return message, the loopback is successful.
- If the destination MP does not respond before the timer expires, the loopback fails.

The LB is similar to the ping operation and therefore is also called MAC PING.

- Link trace (LT)

LT is used to retrieve the MP adjacency and locate faults. The local MEP sends a Link Trace Message which contains the MAC address or MEP ID of the destination MEP and starts a timer.

All MIPs in the same MD as the local MEP on the link receive the Link Trace Message and then forward the message to the destination MEP.

- If the destination MEP returns a Link Trace Reply message to the local MEP, the link trace is successful.

- If the local MEP does not receive the Link Trace Reply message from the destination MEP before the timer expires, the link trace fails.

 NOTE

- The LT function cannot be enabled for links between the local MEP and MIPs.
- If the MEPs are base stations, **CFMMA.BNDVLAN (LTE eNodeB, 5G gNodeB)** is set to **Yes** on one base station, and **CFMMA.BNDVLAN (LTE eNodeB, 5G gNodeB)** is set to **NO** on the other base station, the base stations can check whether the received CC packets carry VLAN tags and adjust the VLAN for sending CC packets accordingly.

In addition, the Link Trace Reply message contains the time to live (TTL) parameter which specifies the position of the MP that responds.

LT is similar to trace route and therefore is also called MAC TRACE.

5.2 Principles (ITU-T Y.1731)

5.2.1 Introduction

Ethernet OAM (ITU-T Y.1731) is an end-to-end Ethernet OAM protocol and an enhancement of Ethernet OAM (IEEE 802.1ag). In addition to the CFM function specified by IEEE 802.1ag, Ethernet OAM (ITU-T Y.1731) also implements the performance management function. The performance management function allows the base station to perform single-ended Ethernet Loss Measurement (ETH-LM), two-way Ethernet Delay Measurement (ETH-DM), and delay variation measurement.

Ethernet OAM (ITU-T Y.1731) is defined by ITU-T Y.1731 (2008). It is supported only on the base station side.

5.2.2 Basic Concepts

The working principle of Ethernet OAM (ITU-T Y.1731) is similar to that of Ethernet OAM (IEEE 802.1ag). For details, see [5.1.2 Basic Concepts](#).

To simplify the implementation, Ethernet OAM (ITU-T Y.1731) combines MD and MA into a maintenance entity group (MEG). In doing so, it also:

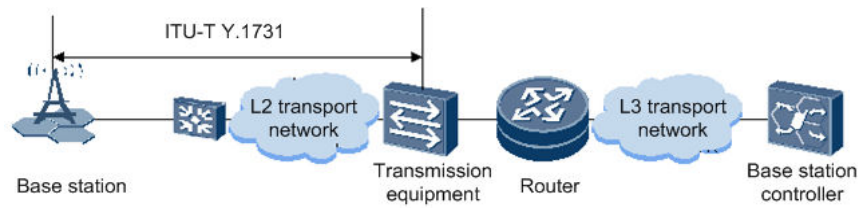
- Replaces the MA ID with MEG ID.
- Replaces MD level with MEG Level.

The definitions of MEPs and MIPs in Ethernet OAM (ITU-T Y.1731) are the same as Ethernet OAM (IEEE 802.1ag).

5.2.3 Application Scenarios

[Figure 5-3](#) shows the network topology with Ethernet OAM (ITU-T Y.1731) enabled. Ethernet OAM (ITU-T Y.1731) is deployed between a base station and transmission equipment and both ends must support ITU-T Y.1731.

Figure 5-3 Network topology with Ethernet OAM (ITU-T Y.1731) enabled



5.2.4 Technical Description

Ethernet OAM (ITU-T Y.1731) inherits the CFM function of Ethernet OAM (IEEE 802.1ag). For details about the CFM function of Ethernet OAM (ITU-T Y.1731), see [5.1.4 Technical Description](#). This section describes the performance management function of Ethernet OAM (ITU-T Y.1731), which is used for single-ended ETH-LM, two-way ETH-DM, and delay variation measurement.

NOTE

For Ethernet OAM (ITU-T Y.1731), ULPU boards support delay measurement with microsecond precision, while other boards support delay measurement with millisecond precision. However, measurement results of all boards are displayed in the unit of microseconds.

Single-ended Ethernet Loss Measurement (ETH-LM)

Single-ended ETH-LM is used for on-demand OAM. LMM and LMR messages contain values of counters for frame loss measurements.

Each MEP on the base station side can measure the following items:

- Local-end frame loss, which is the loss of service frames in the downlink
- Remote-end frame loss, which is the loss of service frames in the uplink

During single-ended ETH-LM, an MEP on the base station side sends an LMM request message to its peer MEP on the transmission equipment and receives an LMR message that carries LMM response information from the peer. The LMM request message contains the value of the local counter TxFCf, TxFCf[tc]. The transmission equipment responds with an LMR message, as shown in [Figure 5-4](#). [Table 5-1](#) describes related counters.

Figure 5-4 Single-ended ETH-LM

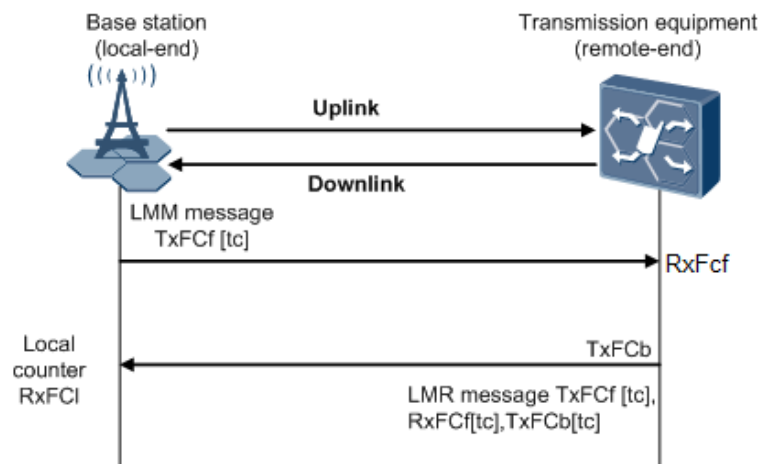


Table 5-1 Counters for single-ended ETH-LM

Counter	Meaning
TxFCf	Local counter at the time of LMM message transmission
RxFCf	Remote counter at the time of LMM message reception
TxFCb	Remote counter at the time of LMR message transmission
RxFCl	Local counter at the time of LMR message reception

The base station compares the values contained in the current and previous LMR messages to calculate local- and remote-end frame loss using the following formulas:

- Remote-end frame loss = (TxFCf[tc] – TxFCf[tp]) – (RxFCf[tc] – RxFCf[tp])
- Local-end frame loss = (TxFCb[tc] – TxFCb[tp]) – (RxFCl[tc] – RxFCl[tp])

Where:

- "c" indicates the current LMR message and "tc" indicates the time the base station received it.
- "p" indicates the previous LMR message and "tp" indicates the time the base station received it.

Two-Way ETH-DM

Two-way ETH-DM measures the delay of DMM/DMR test frames as the delay of the entire Layer 2 network. Figure 5-5 shows the message exchange process. Table 5-2 describes the counters for two-way ETH-DM. The calculation formula is as follows:

Frame delay = (RxTimeb - TxTimeStampf) - (TxTimeStampb - RxTimeStampf)

Figure 5-5 Message exchange for two-way ETH-DM

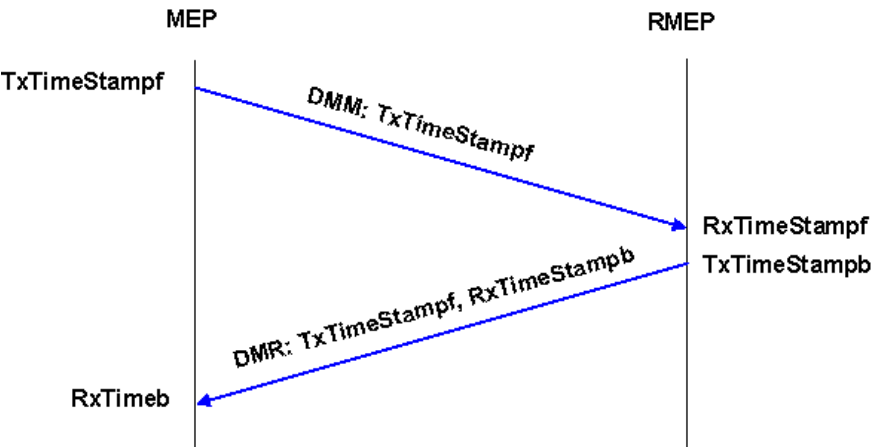


Table 5-2 Counters for two-way ETH-DM

Counter	Meaning
TxTimeSt ampf	Local timestamp at the time of DMM transmission from the local MEP
RxTimeSt ampf	Local timestamp at the time of DMM reception by the remote MEP
TxTimeSt ampb	Local timestamp at the time of DMR transmission from the remote MEP
RxTimeb	Local timestamp at the time of DMR reception by the local MEP

The jitter of a Layer 2 network can be calculated based on the measured two-way delay.

Frame Delay Variation = Frame Delay[tc] - Frame Delay[tp]

 NOTE

tc is the current frame delay of the base station, and tp is the previous frame delay of the base station.

5.3 Network Analysis

5.3.1 Benefits

Ethernet OAM helps operators improve management and maintenance of the Ethernet, guarantee network reliability, and reduce operating expense (OPEX).

5.3.2 Impacts

The impacts between this function and other functions are described in [5.4.2.3 Function Impacts](#).

Ethernet OAM protocols are slow-rate protocols and therefore can provide low data rate functions, such as the ping function. In addition, Ethernet OAM protocols occupy low bandwidth and therefore do not affect the data flow of other services. If the CCM function is enabled, packets must be sent at an interval of at least one second. This is because sending packets at a very short interval affects the performance of transport networks.

5.4 Requirements

5.4.1 Licenses

RAT	Feature ID	Feature Name	Model	Sales Unit
UMTS	WRFD-050425	Ethernet OAM	LQW9ETOAM01	per NodeB
LTE FDD	LOFD-003004	Ethernet OAM	LT1S00EOAM00	per eNodeB
LTE TDD	TDLOFD-003004	Ethernet OAM	LT1ST0EOAM00	per eNodeB
NB-IoT	MLOFD-003004	Ethernet OAM	ML1S00EOAM00	per eNodeB
NR	FOFD-010060	Transmission Network Detection and Reliability Improvement	NR0S0TNDER00	per gNodeB

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists (FDD)*, *License Control Item Lists (CIoT)*, and *License Control Item Lists (TDD)* in eRAN feature documentation, as well as *License Control Item Lists* in 5G RAN feature documentation.

5.4.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.4.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference
GSM	Abis over IP	None	<i>IPv4 Transmission</i>
GSM	A over IP	None	
GSM	Gb over IP	None	
UMTS	IP Transmission Introduction on Iub Interface	None	

5.4.2.2 Mutually Exclusive Functions

Function Name	Function Switch	Reference	Description
Ethernet OAM (IEEE 802.1ag)	None	<i>Ethernet OAM</i>	Ethernet OAM (IEEE 802.1ag) and Ethernet OAM (ITU-T Y.1731) cannot be enabled simultaneously on a base station.
Ethernet OAM (ITU-T Y.1731)	None		

5.4.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
GSM UMTS	Link Aggregation	None	<i>IPv4 Transmission</i>	- Ethernet trunk member ports support Ethernet OAM (ITU-T Y.1731), but Ethernet trunk ports do not. - When Ethernet OAM (ITU-T Y.1731) is enabled for Ethernet trunk ports, single-end packet loss detection is not supported.
LTE FDD	Ethernet Link Aggregation	None		
LTE TDD	Ethernet Link Aggregation (IEEE 802.3ad)	None		
CloT	Ethernet Link Aggregation (IEEE 802.3ad)	None		
NR FDD Low-frequency NR TDD	Transmission Network Detection and Reliability Improvement - Ethernet Link Aggregation	None		
LTE FDD LTE TDD	S1 and X2 over IPv6	None	<i>IPv6 Transmission</i>	Ethernet OAM (IEEE 802.1ag/ITU-T Y.1731) does not support IPv6 transmission.
LTE FDD LTE TDD	S1 over IPv6	None	<i>IPv6 Transmission</i>	Ethernet OAM (IEEE 802.1ag/ITU-T Y.1731) does not support IPv6 transmission.

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequen cy NR TDD	IPv6	None	<i>IPv6 Transmission</i>	Ethernet OAM (IEEE 802.1ag/ITU-T Y.1731) does not support IPv6 transmission.

5.4.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

The following base stations are compatible with this function:

- 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R
- DBS3900 LampSite and DBS5900 LampSite

Boards

RAT	NE	Board	Supported
GSM	BSC	FG2a/FG2c/FG2d/FG2e/ GOUa/GOUc/GOUD/ EXOUa/EXOUb	Yes
GSM	GBTS	GTMUc	Yes
GSM	eGBTS	UMPT	Yes
UMTS	RNC	FG2a/FG2c/FG2d/FG2e/ GOUa/GOUc/GOUD/ EXOUa/EXOUb	Yes
	NodeB	UMPT/UTRPc	Yes
LTE	eNodeB	UMPT/UMDU/ BookBBU5901/ BookBBU5901a/BBU5900C/ UTRPc	Yes
NR	gNodeB	UMPTe/UMPTg/UMPTga/ UMPTga/UMDU/ BookBBU5901/ BookBBU5901a/BBU5900C	Yes

RF Modules

N/A

Transmission Equipment

The transmission equipment directly connected to the peer end must support IEEE 802.1ag or ITU-T Y.1731.

5.4.4 Others

None

5.5 Operation and Maintenance

5.5.1 When to Use

Before Ethernet OAM is enabled, the following conditions must be met:

- Both the base station or base station controller and the transport network are carried on a Layer 2 Ethernet.
- The transmission equipment supports Ethernet OAM.

Ethernet OAM (IEEE 802.1ag) is used to detect and maintain the continuity of end-to-end Ethernet links. This function implements service-based end-to-end continuity check per MD and manages the network segments along the same service flow one by one. In addition, this feature provides link detection and route interlock functions. It is recommended that Ethernet OAM (IEEE 802.1ag) be deployed for a base station or base station controller and transmission equipment that supports IEEE 802.1ag.

If frame loss and delay variation measurements are required for end-to-end Ethernet links, you are advised to deploy Ethernet OAM (ITU-T Y.1731).

The following engineering guidelines for Ethernet OAM are described on a per RAT basis. This is because, in base station co-transmission scenarios, Ethernet OAM is deployed only for a base station that is directly connected to the transport network. Ethernet OAM is deployed only for one RAT of a base station if multiple RATs of the base station are directly connected to the transport network.

5.5.2 Precautions

Ethernet OAM (ITU-T Y.1731) does not support packet loss measurement on an Ethernet trunk. If packets are calculated for the Ethernet trunk, the performance monitoring data is not accurate because the sequence of packets is not saved for the Ethernet trunk.

NOTE

To check packet loss statistics for an Ethernet trunk, you can run the **DSP ETHTRK** command or use the performance counters in the "Measurement of Trunk Performance (TRUNK)" function set.

5.5.3 Data Configuration

5.5.3.1 Data Preparation

GSM

[Table 5-3](#) and [Table 5-4](#) list the data to prepare for activating Ethernet OAM (IEEE 802.1ag/ITU-T Y.1731) on the BSC and GBTS sides, respectively. The data to prepare for activating Ethernet OAM (IEEE 802.1ag) or Ethernet OAM (ITU-T Y.1731) on the eGBTS side is the same as that on the NodeB side. For details, see [Table 5-5](#).

Table 5-3 Data to prepare for activating Ethernet OAM (IEEE 802.1ag) or Ethernet OAM (ITU-T Y.1731) on the BSC side

Parameter Name	Parameter ID	Setting Notes
MD Index	CFMMD.MDIDX	An MD index corresponds to an MD name. The value of the MDIDX parameter for a CFM MD must be unique on the BSC side. The BSC supports a maximum of 7 MDs.
MA Index	CFMMA.MAIDX	An MA index corresponds to an MA name. The value of the MAIDX parameter for a CFM MD must be unique on the BSC side. A CFM MD can contain a maximum of 500 MAs whose indexes range from 0 to 499.
VLAN ID	CFMMA.VLANID	This parameter specifies the VLAN of the MA in the MD. Ethernet CFM supports the creation of MAs for each VLAN within an MD. An MA must correspond to a VLAN ID.
CCM TX Interval	CFMMA.CCMINTERVAL	N/A
VLAN Priority	CFMMA.VLANPRI	In Ethernet networking, VLAN priorities are used to differentiate the priorities of packets at the data link layer. The value 0 indicates the lowest priority and the value 7 indicates the highest priority. The system will preferentially send data packets with a higher priority when links are congested.

Parameter Name	Parameter ID	Setting Notes
MA Index	CFMMEP.MAIDX	An MA index corresponds to an MA name. The value of the MAIDX parameter for a CFM MD must be unique on the BSC side. A CFM MD can contain a maximum of 500 MAs whose indexes range from 0 to 499.
MEP Type	CFMMEP.MEPTYPE	Set this parameter based on the network plan. The MEPTYPE parameter for the local MEP is set to LocalMep or RemoteMep .
MEP ID	CFMMEP.MEPID	Set this parameter based on the network plan. The value of the MEPID parameter for the local MEP must be unique within an MA.
Port Type	CFMMEP.PORTTYPE	N/A
Trunk No.	CFMMEP.TRUNKN	N/A

Table 5-4 Data to prepare for activating Ethernet OAM (IEEE 802.1ag) or Ethernet OAM (ITU-T Y.1731) on the GBTS side

Parameter Name	Parameter ID	Setting Notes
Ethernet OAM Type	BTSETHOAM.OAMTYPE	This parameter can only be set to IEEE_802_1AG(IEEE 802.1ag) or ITU_T_Y1731(ITU-T Y.1731) .
Monitor Policy	BTSETHOAM.MONITORPOLICY	This parameter is valid only when Ethernet OAM Type is set to ITU_T_Y1731(ITU-T Y.1731) . It is recommended that the MONITORPOLICY parameter be set to the default value PORTID_VLANID_VLANPRI(Based on PortID,VLANID and VLAN PRI) .
MD Name	BTSETHMD.MDNAME	Set this parameter based on the network plan. An MD is identified by the name of the MD. MDNAME of a CFM MD must be unique in a base station. This parameter can be set to any value if OAM Type is set to ITU-T Y.1731 .

Parameter Name	Parameter ID	Setting Notes
MD Level	BTSETHMD.MDLEVEL	A maximum of eight CFM MDs can be configured on a base station. The value of the MDLEVEL parameter in the CFM MD ranges from 0 to 7.
MD Name	BTSETHMA.MDNAME	Before adding a CFM MA, add a CFM MD for the CFM MA under the BTSETHMA MO. The value of MDNAME for the CFM MA must be the same as that for the CFM MD. If OAM Type is set to ITU_T_Y1731(ITU-T Y.1731) , replace MEG ID with MA Name . The value of the MA Name parameter must be unique within an MEG.
MA Name	BTSETHMA.MANAME	An MA is identified by the combination of an MD name and an MA name. If OAM Type is set to ITU_T_Y1731(ITU-T Y.1731) , replace MEG ID with MA Name . The value of the MA Name parameter must be unique within an MEG.
VLAN ID	BTSETHMA.VLANID	Do not set VLANID for a CFM MA to a VLAN ID for other CFM MDs.
MD Name	BTSETHLCMEP.MDNAME	Before setting this parameter based on the network plan, configure this parameter for the BTSETHMD MO.
MA Name	BTSETHLCMEP.MANAME	Before setting this parameter based on the network plan, configure this parameter for the BTSETHMA MO.
MEP ID	BTSETHLCMEP.MEPID	Set this parameter based on the network plan. The value of the MEPID parameter for the local MEP must be unique within an MA.
MD Name	BTSETHRMMEP.MDNAME	Ensure that this parameter has been configured in the BTSETHMD MO.
MA Name	BTSETHRMMEP.MANAME	N/A

Parameter Name	Parameter ID	Setting Notes
MEP ID	BTSETHRMMEP.MEPID	Before checking the continuity between a base station and a remote MEP, configure this parameter for the remote MEP on the base station side. In addition, the configured remote MEP must be consistent with that configured on the existing peer system in terms of MD name, MA name, and MEP ID. Otherwise, the remote MEP cannot work properly.

UMTS

Table 5-5 lists the data to prepare for activating Ethernet OAM (IEEE 802.1ag/ITU-T Y.1731) on the NodeB side. The data to prepare for activating this feature on the RNC side is the same as that on the BSC side. For details, see **Table 5-3**.

Table 5-5 Data to prepare for activating Ethernet OAM (IEEE 802.1ag) or Ethernet OAM (ITU-T Y.1731) on the NodeB side

Parameter Name	Parameter ID	Setting Notes
Ethernet OAM Type	GTRANSPARA.OAMTYPE (LTE eNodeB, 5G gNodeB)	GTRANSPARA.OAMTYPE (LTE eNodeB, 5G gNodeB) can only be set to IEEE_802_1AG(IEEE 802.1ag) or ITU_T_Y1731(ITU-T Y.1731) .
Monitor Policy	GTRANSPARA.MONITORPOLICY (LTE eNodeB, 5G gNodeB)	This parameter is valid only when GTRANSPARA.OAMTYPE (LTE eNodeB, 5G gNodeB) is set to ITU_T_Y1731(ITU-T Y.1731) . It is recommended that this parameter be set to the default value.
MD Name	CFMMD.MDNAME (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan. An MD is identified by the name of the MD. MDNAME of a CFM MD must be unique in a base station. This parameter can be set to any value if OAM Type is set to ITU_T_Y1731(ITU-T Y.1731) .

Parameter Name	Parameter ID	Setting Notes
MD Level	CFMMD.LEVEL (<i>LTE eNodeB, 5G gNodeB</i>)	A maximum of eight CFM MDs can be configured on a base station. The value of LEVEL in the CFM MD ranges from 0 to 7. If OAM Type is set to ITU_T_Y1731(ITU-T Y.1731), the value of LEVEL (<i>LTE eNodeB, 5G gNodeB</i>) is used to indicate the MEG level of ITU_T_Y.1731.
MD Name	CFMMA.MDNAME (<i>LTE eNodeB, 5G gNodeB</i>)	Before adding a CFM MA, configure this parameter for the MA in the CFMMD MO. The value of MDNAME (<i>LTE eNodeB, 5G gNodeB</i>) for the CFM MA must be the same as that for the CFM MD. This parameter is invalid when OAM Type is set to ITU_T_Y1731(ITU-T Y.1731).
MA Name	CFMMA.MANAME (<i>LTE eNodeB, 5G gNodeB</i>)	An MA is identified by the combination of an MD name and an MA name. If OAM Type is set to ITU_T_Y1731(ITU-T Y.1731), replace MEG ID with MA Name. The value of the MA Name parameter must be unique within an MEG.
Binding VLAN	CFMMA.BNDVLAN (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the network plan. This parameter specifies whether an MA is bound with a VLAN. MAs must be bound to VLANs for continuity check on Ethernet ports or Ethernet trunks. MAs cannot be bound to VLANs for continuity check on member ports of an Ethernet trunk.
VLAN ID	CFMMA.VLANID (<i>LTE eNodeB, 5G gNodeB</i>)	Set this parameter based on the network plan. If BNDVLAN (<i>LTE eNodeB, 5G gNodeB</i>) is set to YES(Yes) for a CFM MA, do not set VLANID (<i>LTE eNodeB, 5G gNodeB</i>) to a VLAN ID for other CFM MAs.
VLAN Priority	CFMMA.VLANPRI (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter is valid only when BNDVLAN (<i>LTE eNodeB, 5G gNodeB</i>) is set to YES for a CFM MA.

Parameter Name	Parameter ID	Setting Notes
MD Name	CFMMEP.MDNAME (LTE eNodeB, 5G gNodeB)	Before setting this parameter based on the network plan, configure this parameter for the CFMMD MO.
MA Name	CFMMEP.MANAME (LTE eNodeB, 5G gNodeB)	Before setting this parameter based on the network plan, configure this parameter for the CFMMA MO.
MEP ID	CFMMEP.MEPID (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan. The value of the CFMMEP.MEPID (LTE eNodeB, 5G gNodeB) parameter for the local MEP must be unique within an MA. NOTE When the RMV CFMMEP command is being executed to delete a CFM MEP, the CFM MEP cannot be deleted if it is referenced by an ETHDM or ETHLM MO.
MD Name	CFMRMEP.MDNAME (LTE eNodeB, 5G gNodeB)	Ensure that this parameter has been configured in the associated CFMMD MO.
MA Name	CFMRMEP.MANAME (LTE eNodeB, 5G gNodeB)	Ensure that this parameter has been configured in the associated CFMMA MO.
RMEP ID	CFMRMEP.RMEPID (LTE eNodeB, 5G gNodeB)	Before checking the continuity between a base station and a remote MEP, configure this parameter for the remote MEP on the base station side. In addition, the configured remote MEP must be consistent with that configured on the existing peer system in terms of MD name, MA name, and MEP ID. Otherwise, the remote MEP cannot work properly. NOTE Before deleting a CFMRMEP MO, ensure that the corresponding ETHLM and ETHDM MOs do not reference RMEPID of the CFMRMEP MO.

Parameter Name	Parameter ID	Setting Notes
MAC Address	CFMRMEP.MAC (<i>LTE eNodeB, 5G gNodeB</i>)	The MAC address must be a valid unicast address. The MAC address of the remote MEP is obtained from the CCM message. If the MAC address obtained from the CCM message is different from the configured MAC address, the obtained MAC address is used. Set this parameter on the network plan. If multiple remote MEPs are configured in a CFM MA, use different MAC addresses for these remote MEPs. Otherwise, the CFM LB test may be abnormal. NOTE Before deleting a CFMRMEP MO, ensure that the corresponding ETHLM and ETHDM MOs do not reference MAC of the CFMRMEP MO.
Session ID	ETHDM.SESSIONID (<i>LTE eNodeB, 5G gNodeB</i>)	When the ACT ETHDM command is executed to activate ETH-DM, the combination of MDNAME (<i>LTE eNodeB, 5G gNodeB</i>)+MANAME (<i>LTE eNodeB, 5G gNodeB</i>)+MEPID (<i>LTE eNodeB, 5G gNodeB</i>)+RMEPID (<i>LTE eNodeB, 5G gNodeB</i>) must be unique for measurement sessions with the same session ID.
MD Name	ETHDM.MDNAME (<i>LTE eNodeB, 5G gNodeB</i>)	Ensure that this parameter has been configured in the associated CFMMD MO.
MA Name	ETHDM.MANAME (<i>LTE eNodeB, 5G gNodeB</i>)	Ensure that this parameter has been configured in the associated CFMMA MO.
MEP ID	ETHDM.MEPID (<i>LTE eNodeB, 5G gNodeB</i>)	Ensure that this parameter has been configured in the associated CFMMEP MO. When the ACT ETHDM command is executed to activate ETH-DM, one MEP does not support concurrent measurement sessions with multiple RMEPs.
Destination Type	ETHDM.DSTTYPE (<i>LTE eNodeB, 5G gNodeB</i>)	The value RMEP(RMEP) is recommended.

Parameter Name	Parameter ID	Setting Notes
RMEP ID	ETHDM.RMEPID (<i>LTE eNodeB, 5G gNodeB</i>)	If DSTTYPE (<i>LTE eNodeB, 5G gNodeB</i>) is set to RMEP(RMEP) , the CFMRMEP MO corresponding to this parameter must have been configured before the ACT ETHDM command is executed.
MAC Address	ETHDM.MAC (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter is valid only when DSTTYPE (<i>LTE eNodeB, 5G gNodeB</i>) is set to MAC(Remote MAC Address) . Ensure that this parameter has been configured in the associated CFMRMEP MO.
TX Interval	ETHDM.INTERVAL (<i>LTE eNodeB, 5G gNodeB</i>)	N/A
Session ID	ETHLM.SESSIONID (<i>LTE eNodeB, 5G gNodeB</i>)	When the ACT ETHLM command is executed to activate ETH-LM, the combination of MDNAME +MANAME+MEPID+RMEPID must be unique for measurement sessions with the same session ID.
MD Name	ETHLM.MDNAME (<i>LTE eNodeB, 5G gNodeB</i>)	Ensure that this parameter has been configured in the associated CFMMD MO.
MA Name	ETHLM.MANAME (<i>LTE eNodeB, 5G gNodeB</i>)	Ensure that this parameter has been configured in the associated CFMMA MO.
Mode	ETHLM.MODE (<i>LTE eNodeB, 5G gNodeB</i>)	N/A

Parameter Name	Parameter ID	Setting Notes
MEP ID	ETHLM.MEPID (<i>LTE eNodeB, 5G gNodeB</i>)	Ensure that this parameter has been configured in the CFMMEP MO and the referenced local MEP in the CFMMEP MO meets the following conditions: Port Type in the CFMMEP MO is not set to ETHTRK(Ethernet Trunk). - When Port Type in the CFMMEP MO is set to ETH(Ethernet Port), the Cabinet No., Subrack No., Slot No., Subboard Type, and Port No. parameters in the CFMMEP MO cannot be referenced by the ETHTRKLNK MO. When the ACT ETHLM command is executed, one MEP does not support concurrent measurement sessions with multiple RMEPs.
Destination Type	ETHLM.DSTTYPE (<i>LTE eNodeB, 5G gNodeB</i>)	The value RMEP (RMEP) is recommended.
RMEP ID	ETHLM.RMEPID (<i>LTE eNodeB, 5G gNodeB</i>)	If DSTTYPE (<i>LTE eNodeB, 5G gNodeB</i>) is set to RMEP (RMEP), the CFMRMEP MO corresponding to this parameter must have been configured before the ACT ETHLM command is executed.
MAC Address	ETHLM.MAC (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter is valid only when DSTTYPE (<i>LTE eNodeB, 5G gNodeB</i>) is set to MAC(Remote MAC Address). Ensure that this parameter has been configured in the associated CFMRMEP MO.
TX Interval	ETHLM.INTERVAL (<i>LTE eNodeB, 5G gNodeB</i>)	N/A

Scenario-Specific Data Preparation

If route interlock is required based on the CFM CC result, bind the remote MEP to an IP address. The route interlock function is supported only by the NodeB or eNodeB after the CC function in the **CFMMEP** MO is activated in the summary data file. [Table 5-6](#) lists the key parameters to be configured.

Table 5-6 Key parameters to be configured on the NodeB side

Parameter Name	Parameter ID	Setting Notes
MD Name	CFMBNDIP.MDNAME (<i>LTE eNodeB, 5G gNodeB</i>)	Ensure that this parameter has been configured in the associated CFMMD MO.
MA Name	CFMBNDIP.MANAME (<i>LTE eNodeB, 5G gNodeB</i>)	Ensure that this parameter has been configured in the associated CFMMA MO.
RMEP ID	CFMBNDIP.RMEPID (<i>LTE eNodeB, 5G gNodeB</i>)	Ensure that this parameter has been configured in the associated CFMRMEP MO.
IP Address	CFMBNDIP.BNDIP (<i>LTE eNodeB, 5G gNodeB</i>)	This parameter specifies the IP address of the Ethernet port or the Ethernet link aggregation group corresponding to the remote MEP. • Do not perform route interlock on the same active and standby IP routes by using CFM and bidirectional forwarding detection (BFD). • Do not bind two CFMRMEP MOs in an MA to the same IP address. The bound IP address must be different from those of all configured or effective ports (including physical ports and loopback ports) on the local end.

 NOTE

The RNC does not support this function.

LTE/NR

The data preparation for the eNodeB/gNodeB is the same as that for the NodeB. For details, see [Table 5-5](#).

5.5.3.2 Using MML Commands

Before using MML commands, refer to [5.4 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

GSM (on the BSC Side)

```
//Adding an MD
ADD CFMMD: SRN=0, SN=7, MDIDX=0, LEVEL=0, MDNAME="md0";
//Adding an MA
ADD CFMMA: SRN=0, SN=7, MDIDX=0, VLANID=2, MAIDX=0, MANAME="ma0";
//Adding a local MEP
ADD CFMMEP: SRN=0, MAIDX=0, MEPTYPE=LocalMep, MEPID=1, PORTTYPE=ETHER, SN=7, PN=0;
//Adding a remote MEP
ADD CFMMEP: SRN=0, MAIDX=0, MEPTYPE=RemoteMep, MEPID=11, SN=7;
//Activating the CC function
ACT CFMCCMTST: SRN=0, SN=7, MAIDX=0, MEPID=1;
//Starting a CFM LB test
PING MAC: SRN=0,SN=7,MAIDX=0,SRCEPID=1,DESTMEPID=11,Count=5,PKTSize=95,TimeOut=3;
//Starting a CFM LT test
TRC MAC: SRN=0,SN=7,MAIDX=0,SRCEPID=1,DESTMEPID=11,TTL=60, TimeOut=5;
```

GSM (on the GBTS Side)

```
//Setting the OAM type
//Configuring Ethernet OAM (IEEE 802.1ag)
SET BTSETHOAM:IDTYPE=BYID,BTSID=7,OAMTYPE=IEEE_802_1AG;
//Configuring Ethernet OAM (ITU-T Y.1731) with the monitor policy of performance measurement based on
the source MAC address, destination MAC address, and VLAN ID
SET BTSETHOAM:IDTYPE=BYID,BTSID=7,OAMTYPE=ITU_T_Y1731,MONITORPOLICY=SA_DA_VLANID;
//Adding an Ethernet OAM MD
ADD BTSETHMD: IDTYPE=BYID, BTSID=7, MDNAME="CHINA", MDLEVEL=7;
//Adding an Ethernet OAM MA
ADD BTSETHMA: IDTYPE=BYID, BTSID=7, MDNAME="CHINA", MANAME="SHANGHAI", VLANSWITCH=ON,
VLANID=3;
//Adding a local MEP
ADD BTSETHLCMEP: IDTYPE=BYID, BTSID=7, MDNAME="CHINA", MANAME="SHANGHAI", MEPID=1, PN=0;
//Adding a remote MEP
ADD BTSETHRMMEP: IDTYPE=BYID, BTSID=7, MDNAME="CHINA", MANAME="SHANGHAI", MEPID=2;
//Activating the CC function
ACT BTSETHCC: IDTYPE=BYID, BTSID=7, MDNAME="CHINA", MANAME="SHANGHAI", MEPID=1;
//Starting a CFM LB test
PING BTSCFM: IDTYPE=BYID, BTSID=7, MDName=" CHINA ", MANAME=" SHANGHAI ",DSTTYPE=RMEP,
DESTMEPID=2;
//Starting CFM loopback
CFMPING: MDNAME="CHINA", MANAME="SHANGHAI", DSTTYPE=MAC, MAC="00E2-FC00-57B7";
//Starting an LT test
TRC BTSCFM: IDTYPE=BYID, BTSID=7, MDName="CHINA", MANAME="SHANGHAI", DSTTYPE=RMEP,
DESTMEPID=2;
//Starting a single-ended ETH-LM
ACT BTSETHLM: IDTYPE=BYID, MDNAME="CHINA", MANAME="SHANGHAI", DSTTYPE=MAC, MAC="00E2-
FC00-57B7";
//Starting a two-way delay variation measurement
ACT BTSETHDM: IDTYPE=BYID, MDNAME="CHINA", MANAME="SHANGHAI", DSTTYPE=MAC, MAC="00E2-
FC00-57B7";
```

GSM (on the eGBTS Side)

The activation command examples on the eGBTS side are the same as those on the NodeB side. For details, see "UMTS (on the NodeB Side)" in "Activation Command Examples."

UMTS (on the NodeB Side)

```
//Configuring Ethernet OAM (IEEE 802.1ag)
SET ETHOAM: OAMTYPE=IEEE_802_1AG;
//Configuring Ethernet OAM (ITU-T Y.1731) with the monitor policy of performance measurement based on
```

```

the source MAC address, destination MAC address, and VLAN ID
SET ETHOAM: OAMTYPE=ITU_T_Y1731, MONITORPOLICY=SA_DA_VLANID;
//Adding a CFM MD
ADD CFMMD: MDNAME="CHINA", LEVEL=7;
//Adding a CFM MA
ADD CFMMA: MDNAME="CHINA", MANAME="SHANGHAI", SN=6, BNDVLAN=YES, VLANID=3;
//Adding a local MEP
//When GTRANSPARA.TRANS CFGMODE is set to OLD:
ADD CFMMEP: MDNAME="CHINA", MANAME="SHANGHAI", MEPID=1000, SN=6, SBT=BASE_BOARD,
PT=ETH, PN=0;
//When GTRANSPARA.TRANS CFGMODE is set to NEW:
ADD CFMLOCALMEP: MDNAME="CHINA", MANAME="SHANGHAI", MEPID=1000, PT=ETH, PORTID=0;
//Adding a remote MEP
ADD CFMRMEP: MDNAME="CHINA", MANAME="SHANGHAI", RMEPID=2000, MAC="0023-8956-7895";
//Activating the CFM CC function
ACT CFMCC: MDNAME="CHINA", MANAME="SHANGHAI", MEPID=1000;
//Starting CFM loopback
CFMPING: MDNAME="CHINA", MANAME="SHANGHAI", DSTTYPE=MAC, MAC="00E2-FC00-57B7";
//Starting a single-ended ETH-LM
ACT ETHLM: MDNAME="CHINA", MANAME="SHANGHAI", MEPID=1000, DSTTYPE=MAC, MAC="00E2-
FC00-57B7", SESSIONID=1;
//Starting a two-way delay variation measurement
ACT ETHDM: MDNAME="CHINA", MANAME="SHANGHAI", MEPID=1000, DSTTYPE=MAC, MAC="00E2-
FC00-57B7", SESSIONID=1;

```

The activation command examples on the RNC side are the same as those on the BSC side. For details, see "GSM (on the BSC Side)" in "Activation Command Examples."

LTE/NR

The activation command examples on the eNodeB/gNodeB side are the same as those on the NodeB side. For details, see "UMTS (on the NodeB Side)" in "Activation Command Examples."

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

GSM (on the BSC Side)

```

//Deleting a remote MEP
RMV CFMMEP: SRN=0, SN=7, MAIDX=0, MEPID=11;
//Deleting a local MEP
RMV CFMMEP: SRN=0, SN=7, MDIDX=0, MEPID=1;
//Deleting an MA
RMV CFMMA: SRN=0, SN=7, MAIDX=0;
//Deleting an MD
RMV CFMMD: SRN=0, SN=7, MDIDX=0;

```

GSM (on the GBTs Side)

```

//Deleting a remote MEP
RMV BTSETHRMMEP: IDTYPE=BYID, BTSID=7, MDNAME="CHINA", MANAME="SHANGHAI", MEPID=2;
//Deleting a local MEP
RMV BTSETHLCMEP: IDTYPE=BYID, BTSID=7, MDNAME="CHINA", MANAME="SHANGHAI", MEPID=1;
//Deleting an MA
RMV BTSETHMA: IDTYPE=BYID, BTSID=7, MDNAME="CHINA", MANAME="SHANGHAI";
//Deleting an MD
RMV BTSETHMD: IDTYPE=BYID, BTSID=7, MDNAME="CHINA";
//Setting the OAM type
//Configuring Ethernet OAM (IEEE 802.1ag)
SET BTSETHOAM: IDTYPE=BYID, BTSID=7, OAMTYPE="IEEE_802_1ag";
//Configuring Ethernet OAM (ITU-T Y.1731)
SET BTSETHOAM: IDTYPE=BYID, BTSID=7, OAMTYPE="ITU_T_Y1731", MONITORPOLICY="SA_DA_VLANID";

```

The deactivation command examples on the eGBTS side are the same as those on the NodeB side. For details, see "UMTS (on the NodeB Side)" in "Deactivation Command Examples."

UMTS (on the NodeB Side)

```
//Deactivating ETH-DM for Ethernet OAM (ITU-T Y.1731)
DEA ETHDM: SESSIONID=1;
//Deactivating ETH-LM for Ethernet OAM (ITU-T Y.1731)
DEA ETHLM: SESSIONID=1;
//Deleting a remote MEP
RMV CFMRMEP: MDNAME="CHINA", MANAME="SHANGHAI", RMEPID=2000;
//Deleting a local MEP
RMV CFMMEP: MDNAME="CHINA", MANAME="SHANGHAI", MEPID=1000;
//Deleting a CFM MA
RMV CFMMA: MDNAME="CHINA", MANAME="SHANGHAI";
//Deleting a CFM MD
RMV CFMMD: MDNAME="CHINA";
```

The deactivation command examples are the same as those on the BSC side. For details, see "GSM (on the BSC Side)" in "Deactivation Command Examples."

LTE/NR

The deactivation command examples on the eNodeB/gNodeB side are the same as those on the NodeB side. For details, see "UMTS (on the NodeB Side)" in "Deactivation Command Examples."

5.5.4 Activation Verification

GSM

On the BSC Side

Before using the following methods, ensure that the CC function is enabled at both the local and remote ends.

- Method 1
Run the **DSP CFMMEP** command to query the status and MAC address of the remote MEP. If the value of **MAC Address** is that of the remote port, Ethernet OAM (IEEE 802.1ag) has been activated.
- Method 2
Run the **PING MAC** or **TRC MAC** command to start an LB or LT test from the local MEP to the remote MEP.
 - If the ping operation succeeds, the execution is successful.
 - If the ping operation fails, the command execution fails.

On the GBTS Side

The method is the same as method 2 on the BSC side except that the **PING BTSCFM** or **TRC BTSCFM** commands are used to start an LB or LT test.

On the eGBTS Side

The activation verification using MML commands on the eGBTS side is the same as that on the NodeB side. For details, see "On the NodeB Side" in [UMTS](#).

UMTS

On the RNC Side

Before using the following methods, ensure that the CC function is enabled at both the local and remote ends.

- Run the **DSP CFMMEP** command to check the remote MEP status and the MAC address.
If the value of **MEP Type** is **Remote Mep** and that of **MAC Address** is the MAC address of the remote port, Ethernet OAM (IEEE 802.1ag) or Ethernet OAM (ITU-T Y.1731) has been activated.
- Run the **PING MAC** or **TRC MAC** command to start an LB or LT test from the local MEP to the remote MEP.
 - If the ping operation succeeds, the execution is successful.
 - If the ping operation fails, the command execution fails.

On the NodeB Side

- Run the **DSP CFMRMEP** command to check the remote MEP status and the MAC address.
If the value of **Actual MAC Address** is the MAC address of the remote port and that of **Status** is **Normal**, Ethernet OAM (IEEE 802.1ag) or Ethernet OAM (ITU-T Y.1731) has been activated.
- Run the **CFMPING** or **CFMTRACE** command with **Destination Type** set to **RMEP(RMEP)** to start an LB or LT test from the local MEP to the remote MEP.
 - If the ping operation succeeds, the execution is successful.
 - If the ping operation fails, the command execution fails.

NOTE

Before using the preceding methods, ensure that the CC function is enabled at the local and remote ends.

- In scenarios where the remote MEP and MAC address are known, run the **CFMPING** or **CFMTRACE** command. In the command, set **Destination Type** to **MAC(Remote MAC Address)** and specify **MAC Address** to check the connectivity.

NOTE

The CC function is not required for this verification method.

- You can perform the following operations for checking the Ethernet OAM (ITU-T Y.1731) measurement performance:
 - To check the configuration and status of ETH-DM for Ethernet OAM (ITU-T Y.1731):
Run the **LST ETHDM** command to check the configuration of all the ETH-DM sessions.
Run the **DSP ETHDM** command to check delay status on the Ethernet.
 - To check the configuration and status of ETH-LM for Ethernet OAM (ITU-T Y.1731):
Run the **LST ETHLM** command to check the configuration of all the ETH-LM sessions.

Run the **DSP ETHLM** command to check packet loss status on the Ethernet.

- To check the packet loss status on the Ethernet port or Ethernet trunk:
Run the **DSP ETHPORT** command to check the values of **Number of RX Packets**, **Number of RX Bytes**, and **Number of RX CRC Error Packets**.
Run the **DSP ETHTRK** command to check the values of **Number of RX Packets**, **Number of RX Bytes**, **Number of RX Dropped Packets**, and **Number of RX Dropped Bytes**.

LTE/NR

The activation observation procedure on the eNodeB/gNodeB side is the same as that on the NodeB side. For details, see "On the NodeB Side" in [UMTS](#).

5.5.5 Network Monitoring

None

6 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

7 Reference Documents

- *SRAN Networking and Evolution Overview*
- *IPv4 Transmission*
- *IPv6 Transmission*